

ENERGY-EFFICIENT ECG TRANSMISSION

PROJECT REPORT (Phase II)

*Submitted in partial fulfilment of the requirements for the award of
B. Tech Degree in Electronics and Communication Engineering
from the APJ Abdul Kalam Technological University*

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CERTIFICATE

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ABSTRACT

Electrocardiogram (ECG) monitoring is an important requirement in modern wearable healthcare systems, and continuous monitoring is essential for effective health assessment. However, existing ECG transmission systems heavily rely on fixed compression and coding schemes, leading to inefficient energy usage. Since wearable devices operate under strict battery constraints, wireless transmission becomes the major limiting factor in device lifetime. This paper presents an adaptive ECG transmission framework that integrates machine-learning-based classification and channel quality information for intelligent transmission decisions. A Temporal Convolutional Network (TCN) is used to classify ECG segments based on heartbeat characteristics. Experimental results demonstrate that the energy-efficient ECG transmission system achieves 24.2\% energy savings compared to a non-adaptive baseline system. The system demonstrates an effective quality-energy trade-off, achieving substantial energy savings with very little signal distortion, thereby establishing a practical foundation for battery-constrained wearable ECG devices.

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LIST OF SYMBOLS AND ABBREVIATIONS

AWGN	- Additive White Gaussian Noise
BER	- Bit Error Rate
BLE	- Bluetooth Low Energy
CVD	- Cardiovascular Disease
CQM	- Channel Quality Metric
C_R	- Compression Ratio
R_c	- Channel Coding Rate
DWT	- Discrete Wavelet Transform
ECC	- Error-Correcting Code
ECG	- Electrocardiogram
IoT	- Internet of Things
MSE	- Mean Square Error
PRD	- Percent RMS Difference
RPM	- Remote Patient Monitoring
SNR	- Signal-to-Noise Ratio
SQI	- Signal Quality Index
TCN	- Temporal Convolutional Network
WBAN	- Wireless Body Area Network
WBSN	- Wireless Body Sensor Node

CHAPTER 1

INTRODUCTION

The global rise in cardiovascular diseases has created an urgent demand for reliable and continuous cardiac monitoring technologies. Electrocardiography (ECG) is one of the most widely used clinical tools for monitoring the electrical activity of the heart and diagnosing various cardiac abnormalities. Monitoring these waveforms provides valuable information about the physiological condition of the heart.

Recent developments in wearable healthcare technologies have enabled continuous ECG monitoring using compact wearable devices. These devices operate as Wireless Body Sensor Nodes (WBSNs) and are commonly integrated within Internet of Things (IoT) based healthcare systems. In such systems, physiological signals are acquired by wearable sensors and transmitted wirelessly to remote servers or mobile applications where the data can be analyzed by healthcare professionals.

Despite these advantages, wearable ECG monitoring systems face significant challenges due to their limited battery capacity and computational resources. Continuous sensing, processing, and wireless transmission consume substantial energy and limit device lifetime. Among these operations, wireless transmission is widely recognized as the dominant contributor to power consumption.

Traditional ECG monitoring systems typically transmit ECG signals using fixed compression and channel coding parameters. ECG signals contain long periods of normal cardiac activity occasionally interrupted by abnormal events such as arrhythmias. Transmitting all ECG segments with the same transmission parameters results in unnecessary data transmission during normal conditions while failing to provide sufficient protection for clinically important arrhythmic events.

To address these limitations, this project proposes an Adaptive ECG Transmission System that dynamically adjusts compression and channel coding parameters based on the clinical significance of the ECG signal and the quality of the wireless communication channel. The system integrates deep learning based arrhythmia detection with adaptive communication techniques. A Temporal Convolutional Network (TCN) is used to classify ECG segments as either normal or arrhythmic. Based on this classification result and the estimated channel Signal to Noise Ratio (SNR), the system dynamically selects appropriate compression thresholds and channel coding rates.

Normal ECG segments are transmitted with higher compression levels to reduce energy consumption, while arrhythmic segments are transmitted with lower compression and stronger channel protection to preserve diagnostic accuracy. The transmission framework includes ECG signal preprocessing, wavelet-based compression, packetization with error detection, Low Density Parity Check (LDPC) channel coding, and wireless transmission over a simulated fading channel. By intelligently adapting transmission parameters according to signal characteristics and channel conditions, the proposed system improves energy efficiency while maintaining reliable and clinically meaningful ECG transmission.

The development of this system was carried out in stages. Phase I focused on establishing the signal processing and arrhythmia detection foundation required for the adaptive transmission framework, including ECG preprocessing, dataset preparation, and the development and training of a lightweight Temporal Convolutional Network (TCN) classifier for arrhythmia detection. Building upon this foundation, the complete system integrates adaptive compression, LDPC channel coding, wireless transmission modelling, and medical feature validation to evaluate both communication reliability and clinical signal integrity. The results demonstrate that the proposed adaptive framework improves transmission energy efficiency while preserving diagnostically important ECG characteristics.

CHAPTER 2

LITERATURE REVIEW

The development of wearable ECG monitoring systems has attracted significant research interest due to the growing demand for continuous and remote healthcare monitoring. Several studies have investigated methods for improving the energy efficiency, reliability, and accuracy of ECG transmission systems used in wireless body sensor networks.

Alhassan E. Alattar and Saeed Mohsen presented a comprehensive survey of smart wearable healthcare devices. Their work highlighted the importance of energy efficient communication technologies for wearable monitoring systems and identified wireless data transmission as one of the major contributors to energy consumption in such systems. The study also emphasized the use of low power wireless communication protocols such as Bluetooth Low Energy (BLE) for wearable medical devices.

Another important contribution was made by Avrajit Ghosh, Arnab Raha, and Amitava Mukherjee, who investigated energy efficient ECG signal processing techniques using sparse wavelet representations. Their work utilized Discrete Wavelet Transform based signal decomposition and sparse encoding to reduce data redundancy while preserving important ECG features. This approach confirmed that transform based compression techniques are highly effective for biomedical signal transmission in energy constrained environments.

The use of machine learning for ECG analysis has also been widely explored in recent years. Temporal Convolutional Networks (TCNs) have emerged as an effective architecture for analyzing time series biomedical signals such as ECG waveforms. Unlike traditional recurrent neural networks, TCN models use dilated

causal convolutions that allow efficient learning of long range temporal dependencies in ECG signals while maintaining lower computational complexity.

Ingolfsson and colleagues demonstrated that TCN based ECG classifiers can achieve high classification accuracy while requiring significantly fewer parameters and computational operations compared to conventional deep learning models. This makes TCN architectures particularly suitable for deployment on resource constrained wearable devices.

Recent studies have also explored adaptive transmission frameworks that integrate signal processing, machine learning, and communication system design. In these systems, the transmission parameters such as compression ratio and channel coding rate are dynamically adjusted based on the characteristics of the ECG signal and channel conditions. Such cross layer optimization approaches allow efficient utilization of communication resources while ensuring reliable transmission of clinically important data.

The proposed Adaptive ECG Transmission System builds upon these research contributions by combining machine learning based arrhythmia detection with adaptive compression and channel coding strategies. By incorporating both signal awareness and channel awareness into the transmission process, the system aims to achieve a balanced trade off between energy efficiency, communication reliability, and diagnostic signal quality.

CHAPTER 3

SYSTEM MODEL

3.1 PROBLEM STATEMENT

Wearable ECG monitoring systems are widely used for continuous cardiac monitoring in modern healthcare applications. These systems operate as Wireless Body Sensor Nodes (WBSNs) that acquire physiological signals and transmit them to remote monitoring platforms. However, wearable devices operate with limited battery capacity and constrained computational resources, making energy efficient communication a critical design requirement.

In conventional ECG monitoring systems, transmission parameters such as compression ratio and channel coding rate are fixed. As a result, ECG data is transmitted using the same transmission configuration regardless of the signal importance or wireless channel conditions. This approach leads to inefficient energy utilization because a large portion of ECG monitoring time consists of normal cardiac activity that does not require high fidelity transmission.

Another limitation of fixed transmission schemes is that they do not adapt to variations in wireless channel quality. When channel conditions deteriorate due to noise or fading, important ECG data may experience higher error rates, potentially affecting the reliability of medical monitoring systems.

Therefore, there is a need for an intelligent ECG transmission framework that adapts transmission parameters based on both ECG signal significance and wireless channel conditions. The proposed system addresses this by integrating deep learning based ECG classification with adaptive compression and channel coding.

3.2 OBJECTIVES

The main objective of this project is to design and implement an Adaptive ECG Transmission System capable of improving energy efficiency while maintaining reliable and clinically meaningful ECG signal transmission.

The specific objectives of the project are:

1. To develop a lightweight Temporal Convolutional Network (TCN) model for accurate arrhythmia detection in ECG signals.
2. To design a rule based adaptive decision engine that determines appropriate compression and channel coding parameters based on ECG classification results and wireless channel conditions.
3. To implement an efficient ECG compression scheme using Discrete Wavelet Transform based signal decomposition and dead-zone quantization.
4. To integrate adaptive channel coding using Low Density Parity Check (LDPC) codes to improve communication reliability.
5. To evaluate the performance of the proposed system using metrics such as Percentage Root Mean Square Difference (PRD), Mean Square Error (MSE), transmission success rate, and energy consumption.

3.3 SYSTEM FLOW

The proposed Adaptive ECG Transmission System consists of multiple functional blocks that operate sequentially from ECG acquisition to signal reconstruction at the receiver, as shown in Fig. 3.1.

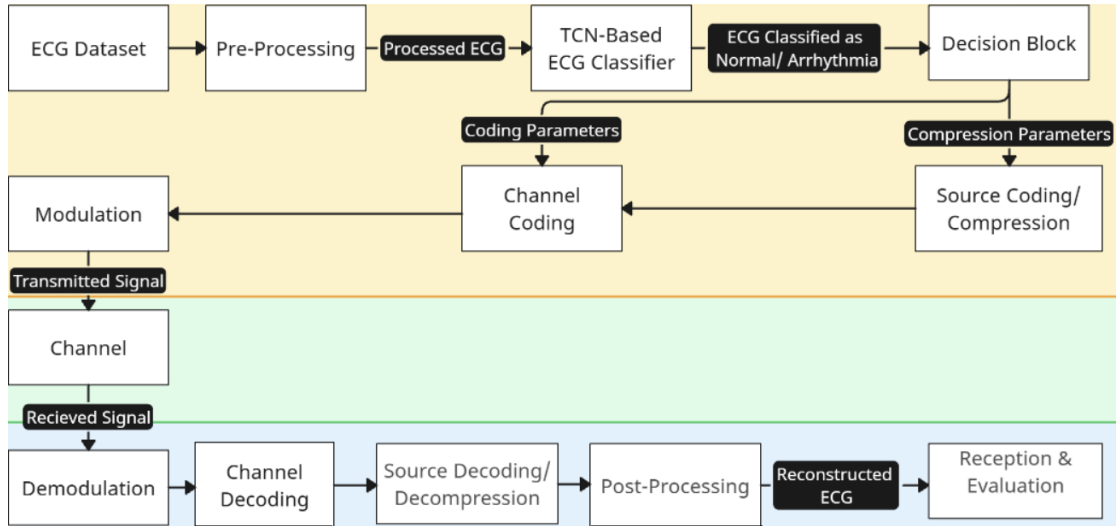


Figure 3.1: Block Diagram of the Adaptive ECG Transmission System

The system begins with ECG data acquisition from a standard database such as the MIT-BIH Arrhythmia Database, which provides annotated ECG signals for analysis. The acquired signal is then passed through a preprocessing block where noise and artifacts such as baseline wander, power line interference, and high-frequency noise are removed. The filtered ECG signal is segmented into fixed-length windows suitable for further processing.

The preprocessed ECG segments are then fed into a Temporal Convolutional Network (TCN)-based classifier. This block classifies each segment as either normal or arrhythmic, thereby determining its clinical importance.

The output of the classifier, along with the channel Signal-to-Noise Ratio (SNR), is provided to the adaptive decision block. This block dynamically selects appropriate compression parameters and LDPC coding rates. Arrhythmic signals are assigned lower compression and stronger error protection to preserve diagnostic features, while normal signals are compressed more aggressively to improve energy efficiency.

Following this, the ECG signal undergoes compression using a Discrete Wavelet Transform (DWT)-based method. The wavelet coefficients are quantized using a dead-zone quantizer and thresholding is applied to remove insignificant components, thereby reducing data size. The compressed data is then packetized, and error detection is incorporated using Cyclic Redundancy Check (CRC) to ensure reliable transmission.

The packetized data is encoded using Low Density Parity Check (LDPC) coding, which introduces redundancy for error correction. The encoded bits are then modulated using Binary Phase Shift Keying (BPSK) and transmitted through a wireless communication channel modeled using Rician fading and Additive White Gaussian Noise (AWGN).

At the receiver, the signal is demodulated using BPSK demodulation, followed by LDPC decoding to correct errors. CRC verification is performed to check packet integrity.

Finally, the ECG signal is reconstructed using inverse wavelet transform. The reconstructed signal is evaluated using performance metrics such as Percentage Root Mean Square Difference (PRD), Signal-to-Noise Ratio (SNR), and clinical feature preservation including R-peak detection and heart rate estimation.

Thus, the system ensures efficient and reliable ECG transmission by adaptively adjusting compression and coding parameters based on signal importance and channel conditions.

CHAPTER 4

METHODOLOGY

The methodology describes the technical implementation of the proposed Adaptive ECG Transmission System. This chapter explains the system architecture, data preparation process, machine learning based arrhythmia detection module, adaptive compression strategy, and the communication framework used for ECG transmission. The design integrates signal processing techniques with machine learning and wireless communication methods in order to achieve energy efficient and reliable ECG monitoring.

4.1 SYSTEM OVERVIEW AND ARCHITECTURE

Modern healthcare monitoring systems increasingly rely on Internet of Things based architectures in which physiological signals are collected using wearable sensors and transmitted wirelessly to remote monitoring platforms. In such systems, the wearable device acts as the sensing layer, while wireless communication technologies form the network layer and cloud or mobile applications form the application layer.

The proposed system introduces an intelligent mechanism to adapt transmission parameters. The ECG signal is preprocessed and analyzed using a Temporal Convolutional Network (TCN) classifier to identify normal and arrhythmic segments.

Based on the classification result and wireless channel conditions, the adaptive decision engine selects appropriate compression and channel coding parameters. These parameters are then applied during the transmission process to achieve energy efficient communication while maintaining signal reliability.

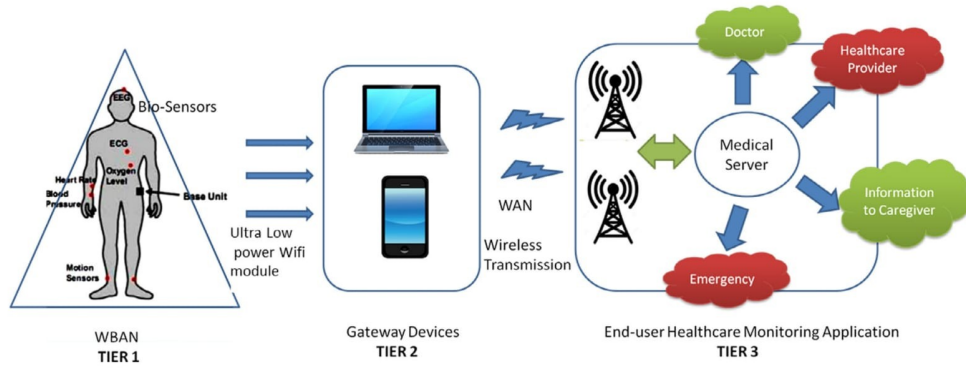


Figure 4.1: General Architecture of an IoT-Based ECG Monitoring System

The architecture shown in Fig. 4.1 illustrates the general framework of an IoT based ECG monitoring system in which wearable sensors collect ECG signals and transmit them through wireless networks to remote healthcare servers.

4.2 DATASETS AND SIGNAL PREPARATION

4.2.1 DATASET EXTRACTION

The ECG data used in this project is obtained from the MIT-BIH Arrhythmia Database, which is one of the most widely used benchmark datasets for ECG signal analysis. The dataset contains 48 half hour ambulatory ECG recordings collected from 47 subjects and sampled at a frequency of 360 Hz.

Each ECG recording in the dataset contains detailed annotations indicating the location and type of each heartbeat. These annotations enable supervised training of machine learning models for arrhythmia detection. In this work, ECG segments are extracted from the dataset and labeled as either normal or arrhythmic based on the provided annotations.

The extracted ECG signals are divided into fixed length segments that serve as input samples for the Temporal Convolutional Network classifier.

4.2.2 ECG SIGNAL VISUALIZATION

ECG signals contain P and T waves and the QRS complex, which represent cardiac electrical activity and help distinguish normal and abnormal rhythms.

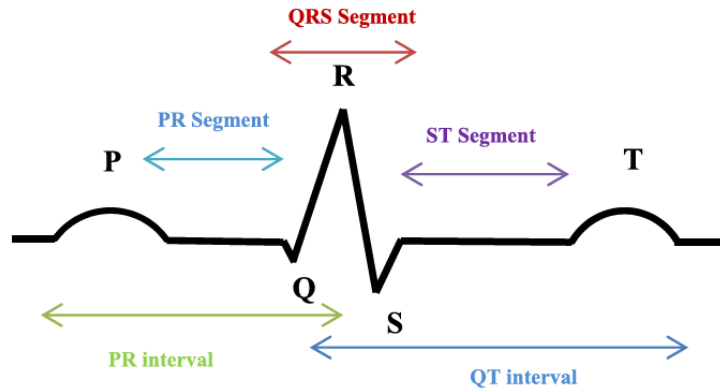


Figure 4.2: General Structure of a Normal ECG Signal

Fig. 4.2 shows the structure of a typical ECG waveform and the different components used for clinical interpretation.

4.2.3 SIGNAL PRE-PROCESSING

Raw ECG signals often contain various types of noise and artifacts such as baseline wander, power line interference, and muscle noise. These disturbances can negatively affect machine learning model performance and must be removed through preprocessing.

The preprocessing stage includes the following steps:

1. Bandpass Filtering: A bandpass filter with a frequency range of approximately 0.5 Hz to 40 Hz is applied to remove baseline drift and high frequency noise.
2. Segmentation: The filtered ECG signal is segmented into fixed length windows suitable for input into the TCN classifier.
3. Normalization: Each ECG segment is normalized using Z-score normalization to ensure consistent signal amplitude and improve model training stability.

These preprocessing steps produce clean ECG segments that can be used effectively for arrhythmia classification.

4.3 FEATURE EXTRACTOR AND DECISION ENGINE

4.3.1 TCN MODEL ARCHITECTURE

The Temporal Convolutional Network serves as the primary feature extraction and classification component in the proposed system. Unlike traditional recurrent neural networks, TCN models use dilated causal convolutions which allow the network to capture long range temporal dependencies in sequential data while maintaining low computational complexity.

The architecture consists of multiple one dimensional convolutional layers with increasing dilation factors. Residual connections are incorporated to improve gradient flow and stabilize the training process.

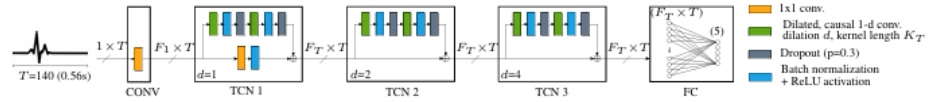


Figure 4.3: Architecture of the Temporal Convolutional Network (TCN)

The trained TCN model analyzes each ECG segment and classifies it as either normal or arrhythmic. The classification output serves as an important input for the adaptive transmission system.

4.3.2 RULE-BASED DECISION ENGINE

The rule based decision engine dynamically selects transmission parameters based on two inputs:

- ECG classification output from the TCN model
- Wireless channel Signal to Noise Ratio (SNR)

When the ECG signal is classified as normal under good channel conditions, higher compression is applied to reduce transmission energy. When arrhythmia is detected or channel quality is poor, lower compression and stronger channel coding are used to preserve signal fidelity and ensure reliable transmission.

4.4 COMPRESSION, CODING, AND CHANNEL

4.4.1 COMPRESSION AND CODING SCHEMES

ECG signal compression is performed using a transform based compression approach. The Discrete Wavelet Transform is used to decompose the ECG signal into different frequency components. This process allows redundant information to be removed while preserving important signal features.

After wavelet decomposition, dead-zone quantization is applied to the wavelet coefficients. The quantization threshold determines the compression level and can be adjusted dynamically according to the classification result.

Channel coding is implemented using Low Density Parity Check codes. LDPC codes provide strong error correction capability while maintaining relatively low computational complexity. Different LDPC coding rates (0.633, 0.604, and 0.504) are selected depending on channel quality in order to balance redundancy and transmission efficiency.

4.4.2 CHANNEL

The wireless communication channel is modeled using a Rician fading channel combined with Additive White Gaussian Noise. The Rician fading model represents the presence of both line of sight and multipath signal components that are commonly observed in wireless body area networks.

Noise introduced by the AWGN channel represents thermal noise and interference present in practical communication systems. The combined Rician fading and AWGN model enables realistic simulation of ECG transmission over wireless links.

4.4.3 RECEIVER

At the receiver, the signal is demodulated to recover the encoded bit stream, followed by LDPC channel decoding to correct transmission errors introduced by the wireless channel.

After channel decoding, packet decoding and integrity verification are carried out using cyclic redundancy check (CRC) to ensure that the received data is error-free. The verified data is then passed to the decompression module.

The decompression stage reconstructs the ECG signal by applying the inverse wavelet transform to the received wavelet coefficients. Finally, the reconstructed ECG signal is evaluated using signal reconstruction metrics such as Mean Square Error (MSE) and Percent Root Mean Square Difference (PRD).

In addition to signal reconstruction metrics, clinical feature validation is performed to ensure that important diagnostic characteristics such as heart rate and R-peak amplitude are preserved after transmission.

4.5 EVALUATION METRICS

The performance of the proposed Adaptive ECG Transmission System is evaluated using a combination of signal quality metrics, communication reliability measures, compression efficiency indicators, and machine learning performance metrics. These metrics are selected to evaluate signal reconstruction quality, transmission reliability, energy efficiency, and classification performance of the proposed adaptive transmission framework.

1. Mean Square Error (MSE)

Mean Square Error measures the difference between the original ECG signal and the reconstructed signal obtained after transmission, decoding, and decompression. A lower MSE indicates better reconstruction accuracy.

$$MSE = \frac{1}{N} \sum_{n=1}^N (x[n] - \hat{x}[n])^2 \quad (4.1)$$

where $x[n]$ represents the original ECG signal sample at index n , and $\hat{x}[n]$ represents the corresponding reconstructed ECG signal sample obtained after transmission, decoding, and decompression.

2. Percentage Root Mean Square Difference (PRD)

PRD is widely used in ECG compression research to evaluate the distortion introduced by compression and transmission.

$$PRD = \sqrt{\frac{\sum_{n=1}^N (x[n] - \hat{x}[n])^2}{\sum_{n=1}^N x[n]^2}} \times 100 \quad (4.2)$$

Lower PRD values indicate better signal reconstruction quality and higher fidelity of the ECG waveform.

3. Energy Consumption per Transmission

Energy efficiency is a key objective of the proposed adaptive transmission system. The energy consumed during transmission is proportional to the number of transmitted bits and the transmission power.

$$E_{tx} = N_{bits} \times E_{bit} \quad (4.3)$$

where N_{bits} is the number of transmitted bits and E_{bit} is the energy required to transmit a single bit.

The adaptive transmission system aims to reduce the overall energy consumption by increasing compression during normal ECG conditions while preserving signal quality during arrhythmic events.

4. Classification Accuracy

The reliability of the TCN based arrhythmia detection module is evaluated using classification accuracy.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (4.4)$$

where TP , TN , FP , and FN represent true positives, true negatives, false positives, and false negatives respectively.

5. Transmission Success Rate

Transmission success rate measures the reliability of the wireless communication link by calculating the percentage of ECG segments that are successfully received and reconstructed without packet failure.

$$Success\ Rate = \frac{N_{success}}{N_{total}} \times 100 \quad (4.5)$$

where $N_{success}$ represents the number of successfully transmitted ECG segments and N_{total} represents the total number of transmitted segments. Higher success rates indicate more reliable communication under different wireless channel conditions.

6. Clinical Feature Preservation

In addition to signal reconstruction metrics, the diagnostic integrity of the transmitted ECG signal is evaluated by comparing important clinical features between the original and reconstructed signals. Key features such as heart rate and R-peak amplitude are analyzed to ensure that clinically relevant information is preserved after transmission.

Maintaining accurate clinical features ensures that the reconstructed ECG signal remains suitable for medical interpretation.

4.6 PERFORMANCE ANALYSIS OF THE SYSTEM

This section presents a theoretical analysis of the proposed ECG compression and transmission system. The analysis focuses on quantization distortion, wireless channel errors, and the expected transmission reliability over a Rician fading channel.

Let $x[n]$, $n = 0, 1, \dots, N - 1$ denote a preprocessed and z-score normalized ECG segment of length $N = 512$ satisfying,

$$\mathbb{E}[x^2[n]] = P_x = 1 \quad (4.6)$$

After compression and quantization, the signal becomes,

$$x_q[n] = x[n] + e_q[n] \quad (4.7)$$

where $e_q[n]$ represents quantization error.

Following LDPC encoding and BPSK modulation, transmitted symbols are,

$$s_k \in \{+1, -1\}, \quad E_s = 1 \quad (4.8)$$

The wireless channel is modeled as a Rician fading channel with additive white Gaussian noise and it is given by,

$$r_k = h_k s_k + w_k \quad (4.9)$$

where h_k is the Rician fading coefficient and w_k is complex Gaussian noise with $\sim \mathcal{CN}(0, N_0)$.

The Rician fading coefficient is expressed as,

$$h_k = \sqrt{\frac{K}{K+1}} e^{j\theta_k} + \sqrt{\frac{1}{K+1}} g_k \quad (4.10)$$

where $g_k \sim \mathcal{CN}(0, 1)$ represents the scattered component and the first term represents the line-of-sight component. The Rician K-factor is given by,

$$K = 10^{K_{\text{dB}}/10} \quad (4.11)$$

For $K_{\text{dB}} = 7$ dB, representing typical Wireless Body Area Network (WBAN) chest-to-belt placement.

After pilot-based channel estimation, zero-forcing equalization, LDPC decoding, and inverse compression, the reconstructed ECG signal is expressed as,

$$\hat{x}[n] = x[n] + e_q[n] + e_c[n] \quad (4.12)$$

where $e_q[n]$ represents quantization error and $e_c[n]$ denotes residual channel-induced distortion.

Assuming statistical independence between quantization and channel errors, the total mean squared error is given by,

$$\text{MSE}_{\text{total}} = \mathbb{E}[(x[n] - \hat{x}[n])^2] = \sigma_q^2 + \sigma_c^2 \quad (4.13)$$

For uniform dead-zone quantization with threshold T , the quantization noise variance is,

$$\sigma_q^2 = \frac{T^2}{12} \quad (4.14)$$

For the adaptive dual-threshold system, the quantization thresholds are chosen. The values of the thresholds are given by,

$$T_{\text{normal}} = 0.28, \quad (4.15)$$

$$T_{\text{arrhythmia}} = 0.12 \quad (4.16)$$

Substituting (10) and (11) in (9),

$$\sigma_{q,N}^2 = \frac{(0.28)^2}{12} = 0.00653 \quad (4.17)$$

$$\sigma_{q,A}^2 = \frac{(0.12)^2}{12} = 0.00120 \quad (4.18)$$

Therefore, the quantization MSE satisfies the values between,

$$0.00120 \leq \sigma_q^2 \leq 0.00653 \quad (4.19)$$

For a balanced dataset (50% normal, 50% arrhythmic segments), the average quantization noise variance is expressed as,

$$\sigma_{q,\text{avg}}^2 = \frac{\sigma_{q,N}^2 + \sigma_{q,A}^2}{2} = 0.00387 \quad (4.20)$$

The channel-induced error σ_c^2 arises from bit errors during wireless transmission. To quantify this component, we first characterize the channel bit error rate, then relate it to signal-level distortion through the LDPC decoder behavior.

From the received signal model in Equation (4), the instantaneous received signal power is given as,

$$P_{\text{signal}} = \mathbb{E}[|h_k s_k|^2] = \mathbb{E}[|h_k|^2] \cdot E_s \quad (4.21)$$

For Rician fading, the average channel gain is given by,

$$\mathbb{E}[|h_k|^2] = \frac{K}{K+1} + \frac{1}{K+1} = 1 \quad (4.22)$$

The noise power is given by the noise variance as,

$$P_{\text{noise}} = \mathbb{E}[|w_k|^2] = N_0 \quad (4.23)$$

Therefore, the average received signal-to-noise ratio per symbol is given by,

$$\bar{\gamma} = \frac{P_{\text{signal}}}{P_{\text{noise}}} = \frac{E_s}{N_0} \quad (4.24)$$

For coherent BPSK demodulation, the instantaneous bit error probability conditioned on the fading coefficient is given by,

$$P_b(\gamma) = Q\left(\sqrt{2\gamma}\right) \quad (4.25)$$

where $\gamma = |h_k|^2 E_s / N_0$ is the instantaneous received SNR and $Q(\cdot)$ is the Gaussian Q-function,

$$Q(x) = \frac{1}{\sqrt{2\pi}} \int_x^\infty e^{-t^2/2} dt \quad (4.26)$$

Averaging over the Rician fading distribution, the average bit error probability is given as,

$$\bar{P}_b = Q\left(\sqrt{\frac{2K\bar{\gamma}}{K+1}}\right) e^{-\frac{\bar{\gamma}}{K+1}} + \frac{1}{2} \left(1 - \sqrt{\frac{K\bar{\gamma}}{K+1+\bar{\gamma}}}\right) \quad (4.27)$$

This expression characterizes the raw channel error rate before LDPC decoding.

The relationship between channel BER and signal-level MSE depends on LDPC decoder performance. Regular LDPC codes with belief propagation decoding exhibit waterfall behavior: above a threshold SNR determined by the code rate, the decoder successfully corrects errors with high probability. For successfully decoded packets validated by CRC-32, residual bit errors are negligible, yielding,

$$\sigma_c^2 \approx 0 \quad \text{for } \bar{\gamma} \geq \gamma_{\text{th}} \quad (4.28)$$

where γ_{th} is the LDPC decoding threshold. For the adaptive LDPC codes (rates 0.504, 0.604, 0.633), density evolution theory predicts threshold SNR values ranging from 4 to 6 dB depending on code rate. When channel conditions exceed this threshold ($\bar{\gamma} \geq \gamma_{\text{th}}$), the system operates in the *quantization-limited regime* where $\text{MSE}_{\text{total}} \approx \sigma_q^2$. Below the threshold, packets fail CRC validation

and are discarded, representing transmission failures rather than contributing to reconstruction error.

The PRD metric is defined as,

$$\text{PRD} = 100 \sqrt{\frac{\sum_{n=0}^{N-1} (x[n] - \hat{x}[n])^2}{\sum_{n=0}^{N-1} x^2[n]}} \quad (4.29)$$

For normalized signals with $P_x = 1$,

$$\text{PRD} = 100 \sqrt{\text{MSE}_{\text{total}}} \quad (4.30)$$

In the quantization-limited regime,

$$\text{PRD} \approx 100 \sqrt{\sigma_q^2} \quad (4.31)$$

For successfully decoded packets at practical WBAN SNR levels ($\text{SNR} \geq 6$ dB), theoretical bounds are derived from quantization analysis as,

MSE Bounds:

$$\text{MSE}_{\min} = \sigma_{q,A}^2 = 0.00120 \quad (\text{pure arrhythmic}) \quad (4.32)$$

$$\text{MSE}_{\text{avg}} = \sigma_{q,\text{avg}}^2 = 0.00387 \quad (\text{balanced}) \quad (4.33)$$

$$\text{MSE}_{\max} = \sigma_{q,N}^2 = 0.00653 \quad (\text{pure normal}) \quad (4.34)$$

Therefore,

$$0.00120 \leq \text{MSE} \leq 0.00653 \quad (4.35)$$

PRD Bounds:

$$\text{PRD}_{\min} = 100\sqrt{0.00120} = 3.46\% \quad (\text{arrhythmic}) \quad (4.36)$$

$$\text{PRD}_{\text{avg}} = 100\sqrt{0.00387} = 6.22\% \quad (\text{balanced}) \quad (4.37)$$

$$\text{PRD}_{\max} = 100\sqrt{0.00653} = 8.08\% \quad (\text{normal}) \quad (4.38)$$

Therefore,

$$3.46\% \leq \text{PRD} \leq 8.08\% \quad (4.39)$$

All theoretical values remain well below the 9% clinical acceptability threshold, ensuring diagnostic validity even under worst-case compression (pure normal signal) conditions. The theoretical average PRD of 6.22% serves as an upper bound for expected reconstruction quality in the quantization-limited regime.

For a frame of N bits with channel bit error rate $\bar{P}_b$, the uncoded frame success probability [11] is,

$$P_{\text{uncoded}} = (1 - \bar{P}_b)^N \quad (4.40)$$

For LDPC-coded transmission, success probability exhibits waterfall behavior. When channel BER is below the code's threshold ($\bar{P}_b < \bar{P}_{b,\text{th}}$), the decoder corrects errors with high probability, yielding near-perfect frame success. The threshold BER values, obtained from density evolution [12] are,

$$\bar{P}_{b,\text{th}}(R = 0.504) \approx 0.015 \quad (4.41)$$

$$\bar{P}_{b,\text{th}}(R = 0.604) \approx 0.010 \quad (4.42)$$

$$\bar{P}_{b,\text{th}}(R = 0.633) \approx 0.008 \quad (4.43)$$

Table 4.6 shows theoretical predictions combining channel BER from Equation (34) with LDPC waterfall behavior. At practical operating SNR (≥ 8 dB), channel BER falls well below LDPC thresholds, resulting in predicted success rates

SNR (dB)	Channel BER	Code Rate	Predicted Success (%)
2	0.052	0.504	5–15
4	0.020	0.504	50–70
6	0.005	0.504	85–95
8	0.0006	0.504	95–100
10	2×10^{-5}	0.604	95–100
12	$< 10^{-5}$	0.604	95–100
15	$< 10^{-6}$	0.633	95–100
20	$< 10^{-8}$	0.633	95–100
25	$< 10^{-10}$	0.633	95–100

Table 4.1: Theoretical Transmission Success Rate vs SNR

exceeding 95%. This confirms operation in the quantization-limited regime where $\sigma_c^2 \approx 0$ and reconstruction quality is determined solely by compression parameters.

CHAPTER 5

RESULTS AND DISCUSSION

This chapter presents the experimental evaluation of the proposed Adaptive ECG Transmission System. The results are discussed following the processing pipeline of the system. The analysis begins with ECG preprocessing and visualization, followed by the performance of the Temporal Convolutional Network (TCN) classifier, transmission reliability of the adaptive communication framework, and finally the quality of the reconstructed ECG signal after wireless transmission.

5.1 ECG SIGNAL PREPROCESSING AND VISUALIZATION

The first stage involves preprocessing raw ECG signals from the MIT-BIH Arrhythmia Database. Bandpass filtering and normalization are applied to remove noise such as baseline wander, high-frequency interference, and motion artifacts while preserving the important morphological characteristics of the ECG waveform.

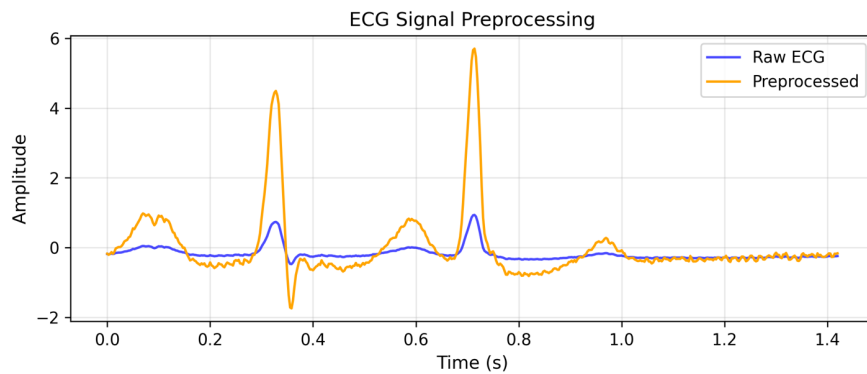


Figure 5.1: Comparison of Raw ECG Signal and Preprocessed ECG Signal

Fig. 5.1 compares raw and preprocessed ECG signals, demonstrating effective noise reduction while preserving essential waveform features.

5.2 TCN MODEL TRAINING AND CLASSIFICATION PERFORMANCE

After preprocessing, ECG signal segments are used to train the Temporal Convolutional Network (TCN) classifier. The classifier distinguishes between normal cardiac rhythms and arrhythmic events so that the adaptive transmission system can assign appropriate compression and channel coding parameters.

5.2.1 Classification Performance Summary

The performance of the trained TCN classifier is evaluated using several classification metrics. The obtained results are summarized in Table 5.1.

Metric	Value
Accuracy	90.0%
Precision	97.62%
Recall	82.0%
F1 Score	89.13%

Table 5.1: TCN Model Classification Performance

These results demonstrate that the classifier effectively identifies arrhythmic ECG segments while maintaining high overall classification performance. The relatively high precision ensures reliable detection of abnormal cardiac rhythms.

5.2.2 Confusion Matrix Analysis

The confusion matrix provides further insight into the classification performance by comparing predicted labels with the true ECG classes.

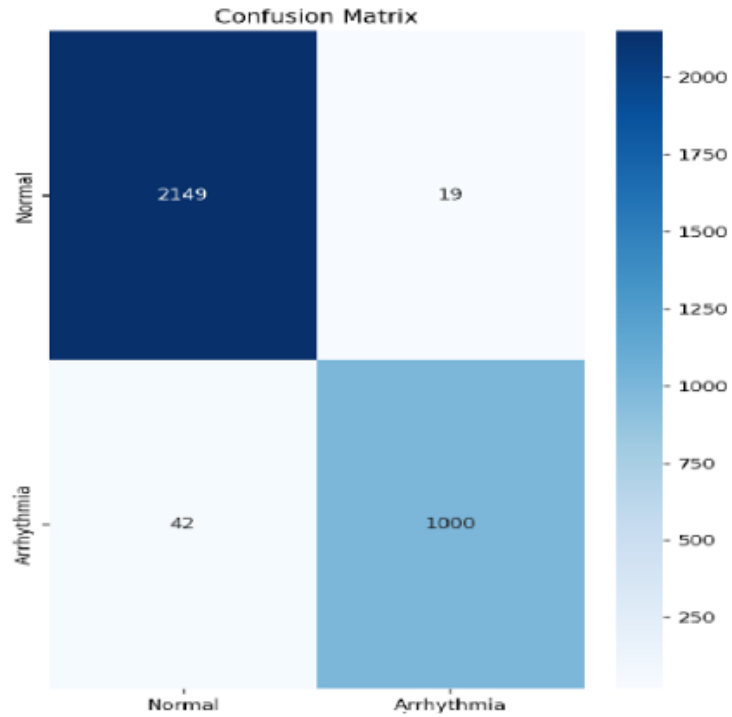


Figure 5.2: Confusion Matrix of the TCN ECG Classifier

As shown in Fig. 5.2, the classifier correctly identifies the majority of ECG segments during the model evaluation phase. The confusion matrix corresponds to the performance obtained on the full testing dataset, where the classifier achieves high accuracy. During the end-to-end system simulation with a smaller balanced subset of ECG segments, the observed classification accuracy is approximately 90%, which reflects the realistic performance of the complete adaptive transmission pipeline.

5.3 TRANSMISSION PERFORMANCE ANALYSIS

Once the ECG segments are classified, the adaptive transmission framework dynamically adjusts compression levels and channel coding parameters before transmitting the data through the wireless channel.

The reliability of wireless transmission is evaluated by measuring the transmission success rate across different Signal-to-Noise Ratio (SNR) values.

The communication channel is modeled using a Rician fading channel combined with Additive White Gaussian Noise to simulate realistic wireless body area network conditions.

For the end-to-end transmission experiments, a balanced subset of 100 ECG segments was randomly selected from the preprocessed dataset. Each segment was processed through the complete adaptive transmission pipeline, including TCN-based classification, adaptive compression, LDPC channel coding, and wireless channel simulation at different SNR values.

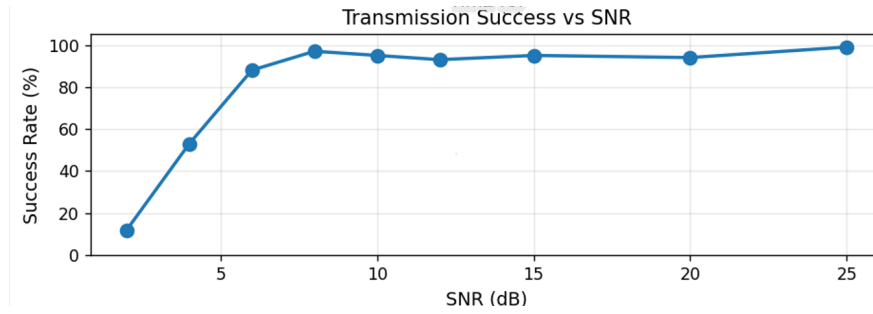


Figure 5.3: Transmission Success Rate versus SNR

Fig. 5.3 shows that the transmission success rate improves significantly as the channel SNR increases. At very low SNR values such as 2 dB, the transmission success rate is relatively low due to severe channel noise. However, when the SNR exceeds approximately 8 dB, the transmission success rate exceeds 95%, demonstrating that the adaptive channel coding mechanism provides reliable communication under moderate wireless channel conditions.

Metric	Adaptive System	Baseline System	Change
Symbols per Beat	6208	8192	-24.2%
Energy (μJ)	6.21	8.19	-24.2%
Success Rate (%)	94.0	96.0	-2.0
PRD (%)	5.37	4.02	+1.35
MSE	0.002886	0.00162	+0.00127

Table 5.2: Performance Comparison between Adaptive and Baseline Transmission

The results show that the adaptive transmission system significantly reduces the number of transmitted symbols and overall energy consumption. Although the

baseline system achieves slightly lower distortion, the adaptive system provides a substantial improvement in energy efficiency while maintaining acceptable signal quality.

5.4 ECG RECONSTRUCTION ANALYSIS

After wireless transmission, the ECG signal is reconstructed at the receiver using channel decoding and inverse compression. The reconstructed signal is compared with the original ECG waveform to evaluate reconstruction quality.

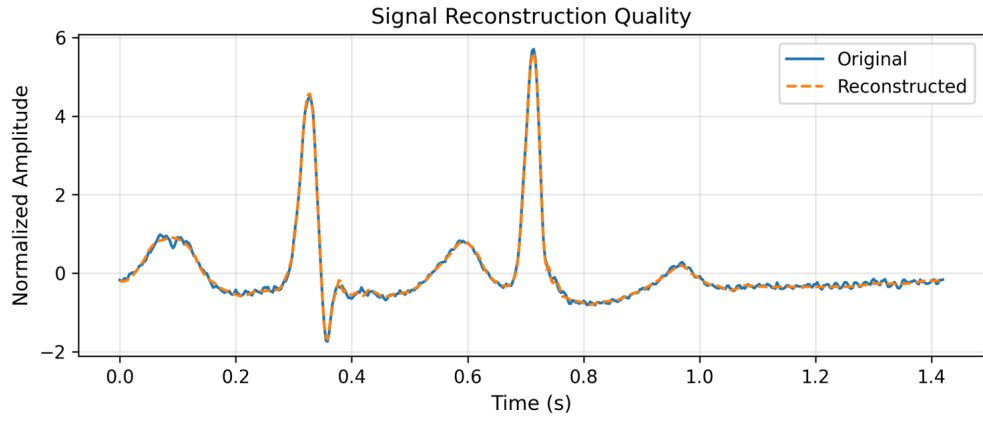


Figure 5.4: Comparison of Original ECG Signal and Reconstructed ECG Signal

As shown in Fig. 5.4, the reconstructed ECG waveform closely follows the original signal. Important ECG features such as the P wave, QRS complex, and T wave remain clearly visible.

Reconstruction accuracy is evaluated using Mean Square Error (MSE) and Percent Root Mean Square Difference (PRD). These metrics correspond to the theoretical performance analysis presented in the methodology chapter.

$$MSE = \frac{1}{N} \sum_{n=1}^N (x[n] - \hat{x}[n])^2 \quad (5.1)$$

$$PRD = \sqrt{\frac{\sum_{n=1}^N (x[n] - \hat{x}[n])^2}{\sum_{n=1}^N x[n]^2}} \times 100 \quad (5.2)$$

The obtained reconstruction metrics indicate high signal fidelity after transmission and reconstruction. The average Mean Square Error (MSE) observed in the experiments is approximately 0.002886, while the Percent Root Mean Square Difference (PRD) is approximately 5.37%. This PRD value remains well below the clinically acceptable threshold of 9%, indicating that the adaptive compression and transmission process introduces only minimal distortion in the ECG signal. These results confirm that the reconstructed ECG signals retain the essential waveform characteristics required for reliable clinical interpretation.

5.5 CLINICAL FEATURE VALIDATION

In addition to signal reconstruction metrics, the clinical integrity of the transmitted ECG signals is evaluated by comparing important diagnostic features between the original and reconstructed signals.

Two clinical parameters are analyzed:

- Heart Rate estimation
- R-peak amplitude

Experimental results show that approximately 95% of the transmitted ECG signals satisfy the clinical validation criteria. The mean heart rate error is approximately 3.28 BPM, while the average R-peak amplitude error is approximately 3.61%.

These results demonstrate that the adaptive transmission system preserves clinically relevant ECG features while achieving significant transmission energy savings.

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 CONCLUSION

This project presented the design and evaluation of an Adaptive ECG Transmission System for energy-efficient wireless cardiac monitoring. The proposed system addresses the limitations of conventional ECG transmission approaches that rely on fixed compression and channel coding parameters, which often lead to unnecessary energy consumption in wearable healthcare devices.

The developed framework integrates signal processing, machine learning, and wireless communication techniques to enable intelligent ECG transmission. A Temporal Convolutional Network (TCN) based classifier is used to analyze ECG signal segments and detect arrhythmic events. The classifier achieved an overall accuracy of approximately 90% during end-to-end system evaluation, enabling reliable identification of clinically significant ECG segments.

Experimental results demonstrate that the adaptive transmission framework significantly improves communication efficiency. The system reduces the number of transmitted symbols by approximately 24% compared to conventional fixed transmission schemes, resulting in a corresponding reduction in transmission energy consumption.

The reconstructed ECG signals maintain high signal fidelity after transmission and reconstruction. The average Mean Square Error (MSE) obtained in the experiments is approximately 0.002886, while the Percent Root Mean Square

Difference (PRD) is approximately 5.37%, which remains well below the clinically acceptable threshold of 9%. In addition, clinical feature validation confirms that approximately 95% of the reconstructed ECG signals preserve important diagnostic parameters such as heart rate and R-peak amplitude.

Overall, the proposed Adaptive ECG Transmission System successfully balances the key design requirements of wearable healthcare systems, namely energy efficiency, communication reliability, and diagnostic signal quality. The integration of deep learning based ECG classification with adaptive compression and channel coding provides a promising approach for future remote cardiac monitoring systems.

6.2 FUTURE WORK

Although the proposed system demonstrates promising performance, several improvements can be explored to enhance its practical applicability.

Future work includes implementing the adaptive ECG transmission framework on embedded hardware platforms such as low-power microcontrollers or wearable sensor nodes for real-time validation in wireless body area networks.

Further research may also investigate lightweight deep learning architectures, such as transformer-based time-series models or hybrid convolutional networks, to improve ECG classification accuracy while maintaining low computational complexity.

In addition, the adaptive decision engine can be extended to incorporate additional parameters such as battery level, network congestion, and patient-specific signal characteristics to enable more intelligent transmission strategies.

Finally, integration with cloud-based healthcare platforms could support large-scale remote patient monitoring and real-time clinical analysis.

REFERENCES

- [1] T. M. Ingolfsson, X. Wang, M. Hersche, A. Burrello, L. Cavigelli and L. Benini, “ECG-TCN: Wearable Cardiac Arrhythmia Detection with a Temporal Convolutional Network,” *Computer Science–Machine Learning*, arXiv.org, Mar. 2021.
- [2] I. S. Fathi, M. A. A. Makhoul, E. Osman and M. A. Ahmed, “An Energy-Efficient Compression Algorithm of ECG Signals in Remote Healthcare Monitoring Systems,” *IEEE Access*, vol. 10, pp. 39130–39144, 2022.
- [3] A. Ghosh, A. Raha and A. Mukherjee, “Energy-Efficient IoT-Health Monitoring System using Approximate Computing,” *Internet of Things*, vol. 9, p. 100166, 2020.
- [4] E. Merdjanovska and A. Rashkovska, “Comprehensive Survey of Computational ECG Analysis: Databases, Methods and Applications,” *Expert Systems with Applications*, vol. 203, 117206, 2022.
- [5] H. S. Pal, A. Kumar, A. Vishwakarma and H.-N. Lee, “Electrocardiogram Signal Compression Using Adaptive Tunable-Q Wavelet Transform and Modified Dead-Zone Quantizer,” *ISA Transactions*, vol. 142, pp. 335–346, 2023.
- [6] M. Blanco-Velasco, F. Cruz-Roldán, J. I. Godino-Llorente, J. Blanco-Velasco, C. Armien-Aparicio and F. López-Ferreras, “On the Use of PRD and CR Parameters for ECG Compression,” *Medical Engineering & Physics*, vol. 27, no. 9, pp. 798–802, 2005.

- [7] R. M. Gray and D. L. Neuhoff, "Quantization," *IEEE Transactions on Information Theory*, vol. 44, no. 6, pp. 2325–2383, Oct. 1998.
- [8] S. Farrokhi, W. Dargie and C. Poellabauer, "Reliable Peak Detection and Feature Extraction for Wireless Electrocardiograms," *Computers in Biology and Medicine*, vol. 185, 109478, 2025.
- [9] X. An, S. Shi, Q. Wang, Y. Yu and Q. Liu, "Research on a Lightweight Arrhythmia Classification Model Based on Knowledge Distillation for Wearable Single-Lead ECG Monitoring Systems," *Sensors*, vol. 24, no. 24, 7896, 2024.
- [10] D. Luong, B. N. Quang, L. Nguyen and M. N. Ngoc, "TinyML-Based System for Cost-Effective ECG Rhythm Classification on Embedded Device," in *2024 International Conference on Advanced Technologies for Communications (ATC)*, 2024.
- [11] J. G. Proakis and M. Salehi, *Digital Communications*, 5th ed., McGraw-Hill, 2008.
- [12] T. Richardson and R. Urbanke, *Modern Coding Theory*, Cambridge University Press, 2008.